

Computer Science & Information Technology

Computer Network - 2

DPP: 1

MAC Sub-Layer

Q1 Match the following.

LAN Standard	Channel Access Protocol
(i) IEEE 802.3	(p) CSMA/CA
(ii) IEEE 802.5	(q) Token
(iii) IEEE 802.11	(r) CSMA/CD

- (A) (i) - (p), (ii) - (q), (iii) - (r)
 (B) (i) - (r), (ii) - (q), (iii) - (p)
 (C) (i) - (r), (ii) - (p), (iii) - (q)
 (D) (i) - (q), (ii) - (r), (iii) - (p)

Q2 Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 10,000 bits. The channel transmission rate is 2 Mbps. Calculate vulnerable time (in millisecond) ?

Q3 Consider a network using the pure ALOHA medium access control protocol, where each frame transmission delay is 2 millisecond. The aggregate number of transmissions across all the nodes is 6000 frames per second. Calculate mean number of frame transmissions per frame time ?

Q4 Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 4 Mbps. The aggregate number of transmissions across all the nodes is 4,000 frames per second. The average number of frames successfully transmitted per second is _____ (rounded to the nearest integer).

Q5 Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 4,000 bits. The channel

transmission rate is 1 Mbps. Calculate vulnerable time (in millisecond) ?

Q6 Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 2,000 bits. The channel transmission rate is 4 Mbps. The aggregate number of transmissions across all the nodes is 2,000 frames per second. The average number of frames successfully transmitted per second is _____. (rounded to the nearest integer)

Q7 Consider a slotted LAN with four number of hosts. Each hosts have data to transmit and transmit with probability 0.2 in every slots. What is the probability (upto three decimal places) that a particular host can transmit in a slot without any collision ?

Q8 Consider a slotted LAN with 30 number of nodes. Each nodes have data to transmit and transmit with probability P in every slots. What is the probability that any node can transmit in a slot without any collision ?

- (A) $29 * P * (1-P)^{28}$
 (B) $29 * P * (1-P)^{29}$
 (C) $30 * P * (1-P)^{29}$
 (D) $30 * P * (1-P)^{30}$

Q9 Consider a slotted LAN with four number of hosts. The hosts have data to transmit and transmit in every slots with the probabilities 0.2, 0.3, 0.4 and 0.5 respectively. What is the probability (upto three decimal places) that any host can transmit in a slot without any collision ?



Q10 Which is NOT a multiple access protocol ?

- (A) ALOHA (B) CSMA
(C) CRC (D) Token

Q11 The minimum frame size (in bytes) required for a CSMA/CD based computer network running at 10 Mbps on a 4 Km cable with a link speed of 2×10^7 meter per second is :

Q12 Determine the maximum length (in kilo-meter) of the cable for transmitting data at a rate of

10 Mbps in an Ethernet LAN with frames of size 2000 bytes. Assume the signal speed in the cable to be 10^7 meter per second.

Q13 Consider A and B are the only two stations on an Ethernet LAN. Each has a steady queue of frames to transmit. Both A and B attempt to transmit a frame, collide and A wins first back-off race. What is the probability (up to three decimal places) that B wins the second back-off race ?



Answer Key

Q1 B
Q2 10~10
Q3 12~12
Q4 541~549
Q5 4~4
Q6 735~741
Q7 0.102~0.103

Q8 C
Q9 0.394~0.394
Q10 C
Q11 500~500
Q12 8~8
Q13 0.125~0.125



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Hints & Solutions

Q1 Text Solution:

LAN Standard

Channel Access Protocol

- | | | |
|-------------------|---|-------------|
| (i) IEEE 802.3 | → | (p) CSMA/CA |
| (ii) IEEE 802.5 | → | (q) Token |
| (iii) IEEE 802.11 | → | (r) CSMA/CD |

Q2 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{10000 \text{ bits}}{2 \text{ Mbps}}$$

$$= \frac{10 \times 10^3 \text{ bits}}{2 \times 10^6 \text{ bits/sec.}} = 5 \text{ ms}$$

Pure ALOHA:

$$\text{Vulnerable time} = 2t_x = 10 \text{ ms}$$

Q3 Text Solution:

$$t_x = 2 \text{ ms}$$

1 sec. 6000 frames

10^3 ms $6 \times 10^3 \text{ frames}$

1 ms 6 frames

$(t_x = 2 \text{ ms})$ 12 frames

Q4 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{1000 \text{ bits}}{4 \text{ Mbps}}$$

$$= \frac{10^3 \text{ bits}}{4 \times 10^6 \text{ bits/sec.}}$$

$$t_x = \frac{1}{4 \times 10^3} \text{ sec.}$$

1 sec. 4000 frames

$$\left(t_x = \frac{1}{4 \times 10^3} \text{ sec.}\right) \Rightarrow \left(\frac{4000}{4 \times 10^3}\right) \text{ frames}$$

$\Rightarrow 1 \text{ frame}$

Mean number of frames transmitted per frame time (G) = 1

Throughput (S) = ?

Pure ALOHA:

$$S = G \times e^{-2G}$$

$$\Rightarrow \frac{1}{e^2} = 0.13533 \text{ to } 0.13717$$

Average number of frames successfully transmitted per sec.

1 sec. (4000 × s) frames

541.34 to 548.69 frames

Q5 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{4000 \text{ bits}}{1 \text{ Mbps}} = \frac{4 \times 10^3 \text{ bits}}{10^6 \text{ bits/sec.}}$$

$$= 4 \text{ ms}$$

Slotted ALOHA:-

$$\text{Vulnerable time} = t_x = 4 \text{ ms}$$

Q6 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{2000 \text{ bits}}{4 \text{ Mbps}}$$

$$= \frac{2 \times 10^3 \text{ bits}}{4 \times 10^6 \text{ bits/sec.}} = \frac{1}{2 \times 10^3} \text{ sec.}$$

1 sec 2000 frames

$$\left(t_x = \frac{1}{2 \times 10^3} \text{ sec.}\right) \Rightarrow \left(\frac{2000}{2 \times 10^3}\right) \text{ frames}$$

$\Rightarrow 1 \text{ frame}$

Mean number of transmission per frame time (G) = 1

Throughput (S) = ?

Slotted ALOHA:

$$S = G \times e^{-G}$$

$$\Rightarrow \frac{1}{e} = 0.367 \text{ to } 0.370$$

Average number of frame transmitted successfully per sec.

1 sec (2000 × s) frames

735.75 to 740.74 frames

Q7 Text Solution:

Number of nodes (N) = 4

Probability (P) = 0.2

Throughput of host = $P \times (1-P)^{(N-1)}$

$$= (0.2) \times (0.8)^{(4-1)}$$

$$= 0.1024$$

Q8 Text Solution:

Number of nodes (N) = 30

Probability = P

Throughput of channel = $N \times P \times (1-P)^{(N-1)}$

$$= 30 \times P \times (1-P)^{29}$$

Q9 Text Solution:

$$\text{Throughput of channel} = [0.2 \times 0.7 \times 0.6 \times 0.5]$$

$$+ [0.8 \times 0.3 \times 0.6 \times 0.5] + [0.8 \times 0.7 \times 0.4 \times 0.5] +$$

$$[0.8 \times 0.7 \times 0.6 \times 0.5] = 0.394$$



Q10 Text Solution:

- CRC

Q11 Text Solution:

CSMA /CD: $t_x \geq 2t_p$

Minimum Frame size = $(2t_p) \times \text{Bandwidth}$

$$= 2 \times \left(\frac{\text{Distance}}{\text{Signal speed}} \right) \times \text{Bandwidth}$$

$$2 \times \left(\frac{4 \text{ KM}}{20 \times 10^7 \text{ m/sec.}} \right) \times 10 \text{ Mbps}$$

$$= 2 \times \left(\frac{4 \times 10^3}{2 \times 10^7 \text{ m/sec.}} \right) \times 10 \times 10^6 \text{ bit/sec}$$

$$\begin{aligned} \text{Minimum frame size} &= 4 \times 10^3 \text{ bits} \\ &= 500 \text{ bytes} \end{aligned}$$

Q12 Text Solution:

Ethernet (CSMA/CD) : $t_x \geq 2t_p$

Maximum distance = $\left(\frac{t_x}{2} \right) \times (\text{Signal speed})$

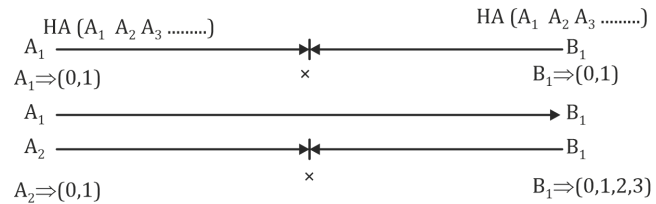
$$= \left(\frac{\text{Frame size} \times \text{Signal speed}}{\text{Bandwidth} \times 2} \right)$$

$$= \left(\frac{2000 \text{ bytes} \times 10^7 \text{ m/sec.}}{10 \text{ Mbps} \times 2} \right)$$

$$\text{Maximum distance} = \frac{2 \times 10^3 \times 8 \text{ bits} \times 10^7 \text{ m/sec.}}{10 \times 10^6 \text{ bits/sec.} \times 2}$$

$$= 8000 \text{ meter}$$

$$= 8 \text{ Km}$$

Q13 Text Solution:

A ₂	B ₁	Result
0	0	Again collided
1	1	
0	1	A win
0	2	
0	3	
1	2	
1	3	B win
1	0	

$$= \frac{1}{8} = \boxed{0.125}$$



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